

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

Objective

This engagement involved reconnoitering scanme.nmap.org and cataloguing the network services exposed by the host using Nmap.

Target: scanme.nmap.org

Tool: Nmap

i. Host Check

Command 1 : nmap -sn scanme.nmap.org

Result:

The scan confirmed that the target host was live and responding to probes.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$
kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:56 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.00061s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 6.34 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:57 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0011s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 886 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
5060/tcp   closed isakmp
993/tcp   closed imaps
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
8081/tcp   closed blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 104.31 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 06:59 EDT
Stats: 0:00:05 elapsed; 0 hosts completed (0 up), 1 undergoing Ping Scan
Ping Scan Timing: About 100.00% done; ETC: 06:59 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0059s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http        Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls?
8080/tcp   closed http-proxy
8081/tcp   closed http-proxy
```

ii. TCP Enumeration

Command 2 : nmap -sS scanme.nmap.org

Result:

The scan output listed each TCP port along with its corresponding state.

```
Session Actions Edit View Help
5061/tcp open  sip-tls?
8080/tcp closed http-proxy
8081/tcp closed blackice-icecap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 98.98 seconds

kali@kali:~$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:01 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0080s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 986 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
993/tcp   closed imaps
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
8081/tcp   closed blackice-icecap
8085/tcp   closed unknown
OS fingerprint not ideal because: maxTimingRatio (3.404000e+00) is greater than 1.4
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 88.24 seconds

kali@kali:~$ nmap -sS scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 07:03 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.018s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
5060/tcp   open  sip
5061/tcp   open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 23.26 seconds

kali@kali:~$
```

Port Details:

TCP Port	Service	Status
22	SSH	Open
80	HTTP	Open

iii. Identify Running Services , Identify OS, Default NSE Scripts

Command 3 : nmap -sV scanme.nmap.org
Command 4 : sudo nmap -O scanme.nmap.org
Command 5 : nmap -sC scanme.nmap.org

Result:

- Nmap resolved the service running behind each open port.
- OS fingerprinting was carried out to estimate the likely operating-system family of the target.
- The built-in NSE default scripts pulled additional detail about the active services.

Security Observations

The enumeration results reveal which services this host exposes to the wider Internet. An adversary could use this information for reconnaissance, for hunting known vulnerabilities, or for identifying accounts to target with credential-based attacks.

Security Measures

- Turn off any service that isn't actually needed.
- Limit who can reach remote-administration interfaces.
- Keep security patches current.
- Build firewall rules around the principle of least privilege.
- Watch for and flag unusual connection attempts.

Final Result

Overall, the Nmap scan gave a clear picture of the target's exposure - the services reachable, the ports they run on, version details, and likely OS - reinforcing why publicly facing systems should be pared down to the minimum needed.

2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

Objective :

This exercise involved running a full Nmap enumeration against the Metasploitable 2 host to catalogue its open ports, active services, service versions, operating system, and associated security exposure.

Target : Metasploitable 2

IP Address : 192.168.56.102

Network : Host-Only Adapter

Tool : Nmap

Procedure

- Set up Metasploitable 2 as a VirtualBox VM.
- Linked the Kali attack VM to Metasploitable over a Host-Only network.
- Confirmed the target's IP address as 192.168.56.102.
- Checked reachability using: `ping -c 4 192.168.56.102`
- Verified the host was live using: `nmap -sn 192.168.56.102`
- Ran a full Nmap enumeration to pull ports, services, versions, OS details, and NSE script output.

Commands and Results

A. Host Discovery

Command : `nmap -sn 192.168.56.102`

Result : The target responded and was confirmed live.

B. TCP Port Scanning

Command : `sudo nmap -sS -p- 192.168.56.102`

Result : Nmap returned the list of open TCP ports on the target.

```
kali@kali: ~$ ping -c 4 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.79 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.72 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=1.62 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.31 ms

--- 192.168.56.102 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 303ms
rtt min/avg/max/ndev = 1.723/2.718/3.789/0.773 ms

kali@kali: ~$ nmap -sn 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:20 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).
MAC Address: 08:00:27:73:FB:18 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds

kali@kali: ~$ sudo nmap -sS -p- 192.168.56.102
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:21 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0011s latency).
Not shown: 65503 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  x11
6667/tcp  open  irc
6697/tcp  open  ircs-u
```

C. Service Version Detection

Command : `sudo nmap -sV 192.168.56.102`

Result : Each service was matched to its running version.

```
kali@kali: ~$ sudo nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:21 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0006s latency).
Not shown: 65507 closed tcp ports (reset)
PORT      STATE SERVICE             VERSION
21/tcp    open  ftp                 vsftpd 2.3.4
22/tcp    open  ssh                 OpenSSH 8.7p1 Debian bullseye (protocol 2.0)
23/tcp    open  telnet              Linux telnetd
25/tcp    open  smtp                Postfix smtpd
53/tcp    open  domain              ISC BIND 9.4.2
80/tcp    open  http                Apache/2.4.18 ((Ubuntu))
111/tcp   open  rpcbind             2 (RPC #100000)
139/tcp   open  netbios-ssn         Samba smbd 3.8.0 - 4.0 (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn         Samba smbd 3.8.0 - 4.0 (workgroup: WORKGROUP)
512/tcp   open  exec                netkit-rsh rshd
513/tcp   open  login               OpenSSH or Solaris rlogind
514/tcp   open  shell               Netkit rshd
1099/tcp  open  java-rmi            GNU Classpath gmiRegistry
1524/tcp  open  ingreslock          Metasploitable root shell
2049/tcp  open  nfs                 2-4 (RPC #100000)
2121/tcp  open  ftp                 ProFTPD 1.3.5
3306/tcp  open  mysql               MySQL 5.6.51a-Debian
3632/tcp  open  distcc              Distcc 3.10.0 - 3.1.7
5432/tcp  open  postgresql          PostgreSQL 9.6.18 - 9.6.7
5900/tcp  open  vnc                  vnc (protocol 3.3)
6000/tcp  open  x11                  (access denied)
6667/tcp  open  irc                  UnrealIRCd
6697/tcp  open  ircs                 Apache/2.4.18 (Ubuntu)
8188/tcp  open  http                Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:73:FB:18 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain, ip: metasploitable.localdomain, OS: Linux, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds

kali@kali: ~$ sudo nmap -sV 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 08:22 EDT
Nmap scan report for 192.168.56.102
Host is up (0.0006s latency).
Not shown: 65507 closed tcp ports (reset)
PORT      STATE SERVICE             VERSION
21/tcp    open  ftp                 vsftpd 2.3.4
22/tcp    open  ssh                 OpenSSH 8.7p1 Debian bullseye (protocol 2.0)
23/tcp    open  telnet              Linux telnetd
25/tcp    open  smtp                Postfix smtpd
53/tcp    open  domain              ISC BIND 9.4.2
80/tcp    open  http                Apache/2.4.18 ((Ubuntu))
111/tcp   open  rpcbind             2 (RPC #100000)
139/tcp   open  netbios-ssn         Samba smbd 3.8.0 - 4.0 (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn         Samba smbd 3.8.0 - 4.0 (workgroup: WORKGROUP)
512/tcp   open  exec                netkit-rsh rshd
513/tcp   open  login               OpenSSH or Solaris rlogind
514/tcp   open  shell               Netkit rshd
1099/tcp  open  java-rmi            GNU Classpath gmiRegistry
1524/tcp  open  ingreslock          Metasploitable root shell
2049/tcp  open  nfs                 2-4 (RPC #100000)
2121/tcp  open  ftp                 ProFTPD 1.3.5
3306/tcp  open  mysql               MySQL 5.6.51a-Debian
3632/tcp  open  distcc              Distcc 3.10.0 - 3.1.7
5432/tcp  open  postgresql          PostgreSQL 9.6.18 - 9.6.7
5900/tcp  open  vnc                  vnc (protocol 3.3)
6000/tcp  open  x11                  (access denied)
6667/tcp  open  irc                  UnrealIRCd
6697/tcp  open  ircs                 Apache/2.4.18 (Ubuntu)
8188/tcp  open  http                Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:73:FB:18 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain, ip: metasploitable.localdomain, OS: Linux, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 13.27 seconds
```



```

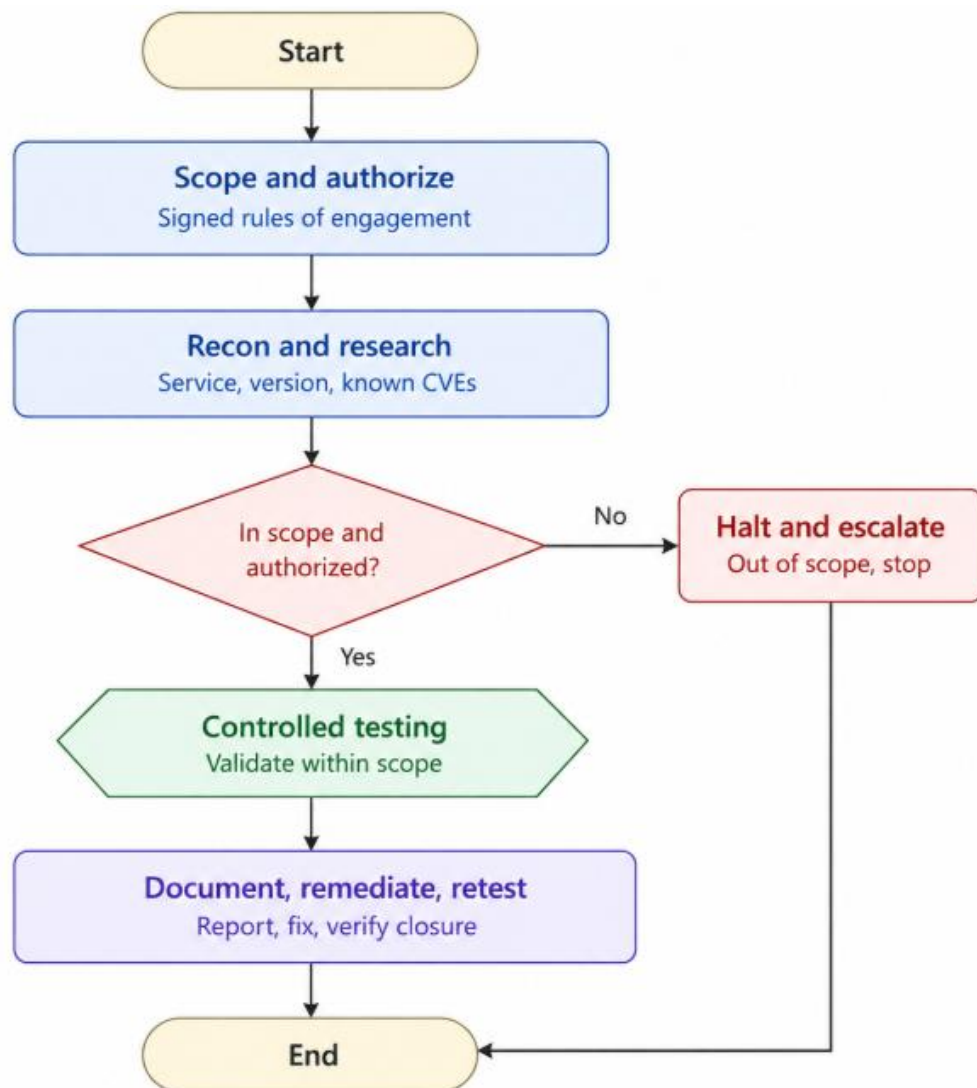
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:73:fb:10
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe73:fb10/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:57 errors:0 dropped:0 overruns:0 frame:0
          TX packets:72 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:8714 (8.5 KB)  TX bytes:6464 (6.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:91 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19301 (18.8 KB)  TX bytes:19301 (18.8 KB)

msfadmin@metasploitable:~$

```

Security Attack Analysis :



Service Risk Analysis :

- FTP - file-transfer service - High risk
- Telnet - unencrypted remote login - High risk
- HTTP - web service - High risk
- Database service - stores/serves data - High risk
- SSH - encrypted remote administration - Medium risk
- SMTP - mail relay - Medium risk
- DNS - name resolution - Medium risk

Recommendations

- Turn off services that aren't needed.
- Swap Telnet out for SSH.
- Keep database services off the general network.
- Patch all software on a regular cadence.
- Enforce strong authentication.
- Tighten firewall rules.

Conclusion

The Nmap enumeration successfully mapped the live host, its open ports, active services and versions, and the underlying OS of the Metasploitable 2 machine. The services uncovered span a range of risk levels and mark out the likely entry points a penetration tester would prioritize during an authorized engagement.

3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks

Objective

This assessment set out to gauge the security posture of the Windows workstation at 192.168.56.102 ahead of connecting it to production. Nmap was used to surface open ports, active services, file-sharing protocols, remote-management interfaces, and OS details.

Target: 192.168.56.102

Tool: Nmap

i. Commands :

Scanning	Commands
Host Availability Scan	ping -c 4 192.168.56.102
TCP Port Scan	sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102
Service Version Scan	sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102
Windows Services Scan	sudo nmap -Pn -sS -sV -p 135,139,445,3389,5985,5986,22 192.168.56.102
OS Detection Scan	sudo nmap -Pn -O --osscan-guess 192.168.56.102
UDP Service Scan	sudo nmap -Pn -sU -sV -p 5060,5061 192.168.56.102

```
kali@kali: ~  
Session Actions Edit View Help  
Host is up.  
  
PORT      STATE SERVICE VERSION  
22/tcp    filtered ssh  
135/tcp   filtered msrpc  
139/tcp   filtered netbios-ssn  
445/tcp   filtered microsoft-ds  
3389/tcp  filtered ms-wbt-server  
5985/tcp  filtered wsdman  
5986/tcp  filtered wsman  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 5.52 seconds  
  
[kali@kali:~]$ sudo nmap -Pn -O --oscan-guess 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:09 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0061s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter[bridge]general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oraclevirtualbox:cpw/adammy.gasparyovskislirp:cpw/qemu/qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 81.47 seconds  
  
[kali@kali:~]$ sudo nmap -Pn -sU -sV -p 5060,5061 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:11 -0400  
Nmap scan report for 192.168.56.102  
Host is up.  
  
PORT      STATE SERVICE VERSION  
5060/udp  open/filtered sip  
5061/udp  open/filtered sip-tls  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 105.51 seconds  
  
[kali@kali:~]$
```

```
kali@kali: ~  
Session Actions Edit View Help  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.88% done  
  
[kali@kali:~]$ sudo nmap -Pn -sS --top-ports 1000 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:03 -0400  
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 3.50% done; ETC: 08:06 (0:03:41 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.00% done; ETC: 08:06 (0:03:36 remaining)  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.50% done; ETC: 08:06 (0:03:32 remaining)  
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.00% done; ETC: 08:06 (0:03:29 remaining)  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.50% done; ETC: 08:06 (0:03:26 remaining)  
Nmap scan report for 192.168.56.102  
Host is up (0.0086s latency).  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
  
Nmap done: 1 IP address (1 host up) scanned in 69.45 seconds  
  
[kali@kali:~]$ sudo nmap -Pn -sV -sC -p 5060,5061 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:04 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.024s latency).  
  
PORT      STATE SERVICE VERSION  
5060/tcp  open  sip? 5061/tcp  open  sip-tls?  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 147.42 seconds  
  
[kali@kali:~]$ sudo nmap -Pn -sS -sV -p 135,139,445,3389,5985,5986,22 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:07 -0400  
Nmap scan report for 192.168.56.102  
Host is up.  
  
PORT      STATE SERVICE VERSION  
22/tcp    filtered ssh  
135/tcp   filtered msrpc
```

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
[kali@kali:~]$  
[kali@kali:~]$ ping -c 4 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
ping: Warning: invalid tv_usec -88005735319229642 us  
ping: Warning: time of day goes back (-87967480382333 s), taking countermeasures  
ping: Warning: invalid tv_usec -88005735319229382 us  
ping: Warning: time of day goes back (-87967480382333 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 83 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319229283 us  
ping: Warning: time of day goes back (-87967480382333 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 83 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319229278 us  
ping: Warning: time of day goes back (-87967480382333 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x18 but was 0xc0  
#16  c0 a8 38 66 8 0 d1 83 4a 23 0 3 c9 bc 79 6a 0 0 0 0 d8 5b 2 0 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
  
--- 192.168.56.102 ping statistics ---  
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 304ms  
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4  
  
[kali@kali:~]$  
[kali@kali:~]$ sudo nmap -Pn -sS -p- 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 07:58 -0400  
Stats: 0:00:14 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 0.88% done  
  
[kali@kali:~]$ sudo nmap -Pn -sS --top-ports 1000 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 08:03 -0400  
Stats: 0:00:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 3.50% done; ETC: 08:06 (0:03:41 remaining)  
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.00% done; ETC: 08:06 (0:03:36 remaining)  
Stats: 0:00:12 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 4.50% done; ETC: 08:06 (0:03:32 remaining)  
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 5.00% done; ETC: 08:06 (0:03:29 remaining)
```

ii. Security Analysis

- Ports 5060 and 5061 stand out as the primary exposed services. Where SIP/VoIP isn't a requirement for this workstation, keeping them open needlessly widens the attack surface.
- SMB, RDP, WinRM, SSH, and NetBIOS all came back filtered, cutting off the more common paths attackers use against file-sharing and remote-management services.
- An attacker could probe the exposed SIP services for reconnaissance, and if misconfigured or vulnerable, potentially escalate that into unauthorized access or further attacks.

iii. Recommendations

- Turn off SIP services if they aren't in active use.
- Where SIP is needed, restrict it to approved systems only.
- Keep SIP software patched and properly hardened.
- Keep firewall rules blocking SMB, RDP, WinRM, and other services that aren't required.
- Disable Windows services that serve no purpose here.
- Apply network segmentation and strong authentication.
- Re-run the Nmap scan once hardening is complete to confirm the changes took effect.

iv. Conclusion

This workstation's exposed attack surface is fairly narrow, limited mainly to the SIP-related ports 5060 and 5061. SMB, RDP, WinRM, SSH, and NetBIOS were all filtered. The remaining SIP exposure should be disabled or tightly restricted to keep the system aligned with least privilege before it goes into production.

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure.

Objective

This assessment enumerates the OWASP Juice Shop server at 192.168.56.103 using Nmap, covering its network exposure, open ports, active services, service versions, operating system, and web-facing components.

Tools Used

Kali Linux, Nmap, and the OWASP Juice Shop application

Target IP: 192.168.56.103

Enumeration Procedure

- i. **Host Discovery : nmap -sn 192.168.56.103**

Result : Confirms whether the target server is up.

- ii. **Port Scanning : nmap -p- 192.168.56.103**

Result : Sweeps the full 65,535 TCP port range and flags which services are reachable.

iii. Service and Version Detection : nmap -sV 192.168.56.103

Result : Reveals the services bound to each open port along with their versions.

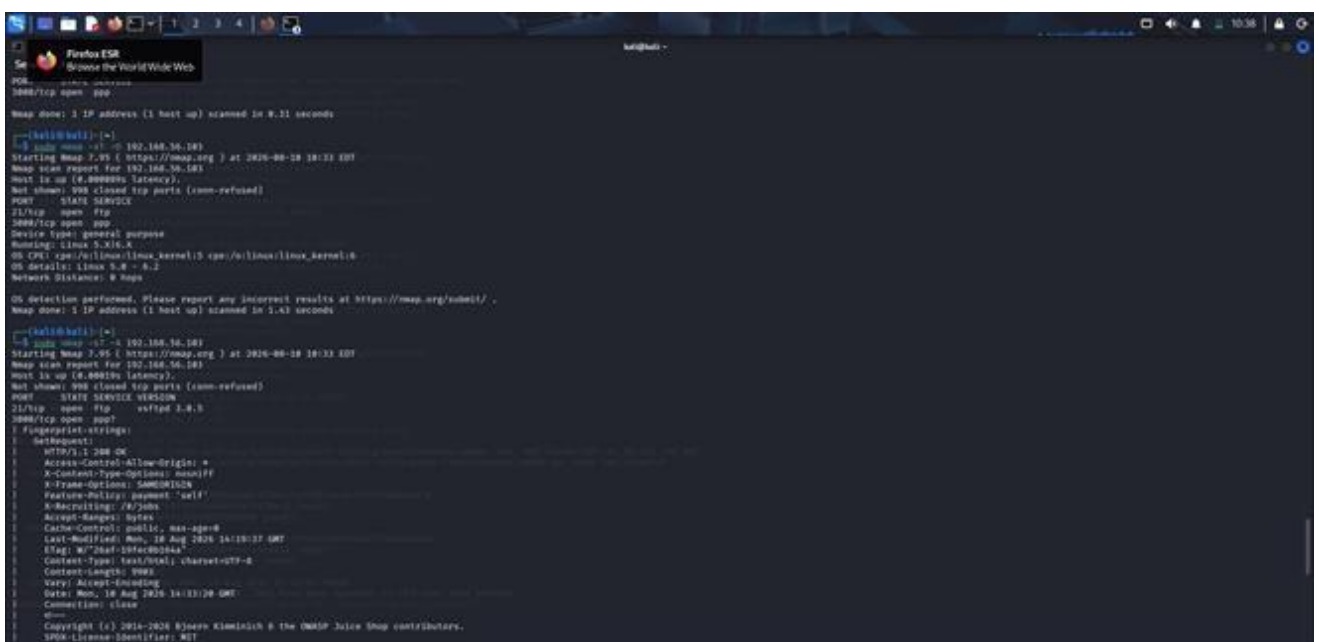
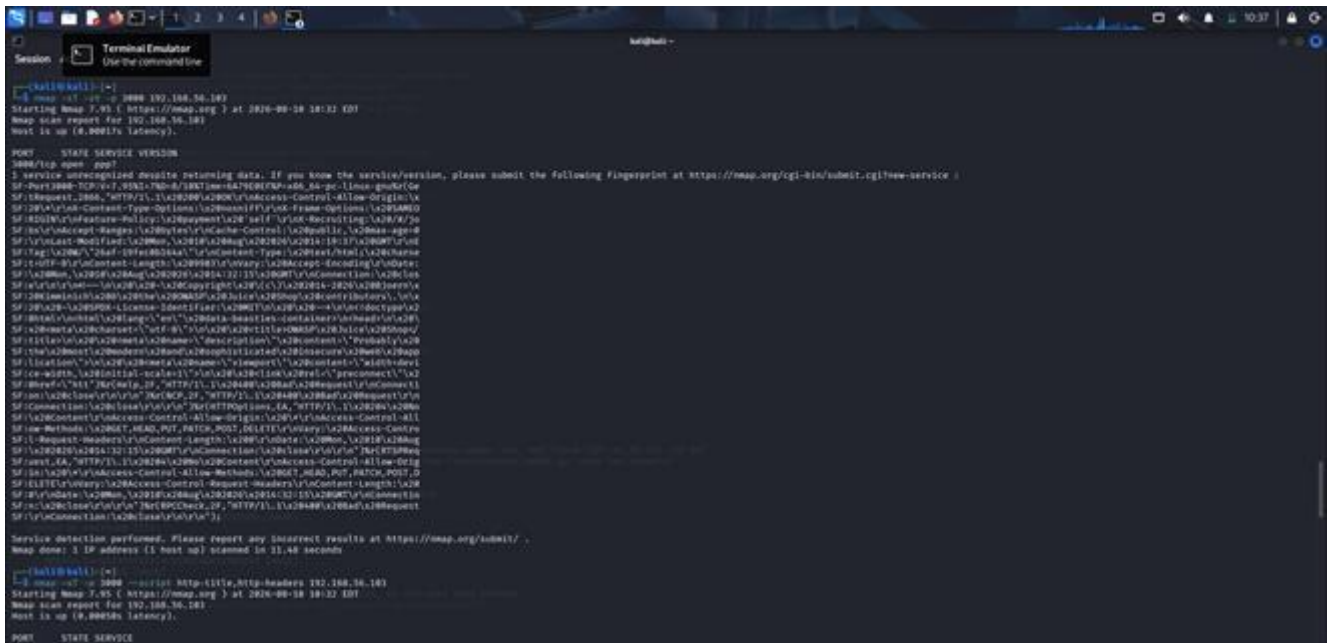
iv. OS Detection : sudo nmap -O 192.168.56.103

Result : Attempts to fingerprint the target server's operating system.

```
kali@kali: ~  
$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:  
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms  
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.096 ms  
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms  
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms  
--- 192.168.56.103 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3062ms  
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms  
  
kali@kali: ~  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up.  
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds  
  
kali@kali: ~  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000017s latency).  
Not shown: 65531 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
3000/tcp  filtered ppp  
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds  
  
kali@kali: ~  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000050s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
3000/tcp  filtered ppp  
Service Info: OS: Unix  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds  
  
kali@kali: ~  
$ sudo docker port juice-shop
```

```
kali@kali: ~  
$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data:  
64 bytes from 192.168.56.103: icmp_seq=1 ttl=64 time=0.174 ms  
64 bytes from 192.168.56.103: icmp_seq=2 ttl=64 time=0.096 ms  
64 bytes from 192.168.56.103: icmp_seq=3 ttl=64 time=0.063 ms  
64 bytes from 192.168.56.103: icmp_seq=4 ttl=64 time=0.038 ms  
--- 192.168.56.103 ping statistics ---  
4 packets transmitted, 4 received, 0% packet loss, time 3062ms  
rtt min/avg/max/mdev = 0.038/0.092/0.174/0.051 ms  
  
kali@kali: ~  
$ nmap -sn 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up.  
Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds  
  
kali@kali: ~  
$ nmap -p- 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:23 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000017s latency).  
Not shown: 65531 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
3000/tcp  filtered ppp  
Nmap done: 1 IP address (1 host up) scanned in 2.45 seconds  
  
kali@kali: ~  
$ nmap -sV 192.168.56.103  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-10 10:24 EDT  
Nmap scan report for 192.168.56.103  
Host is up (0.000050s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
3000/tcp  filtered ppp  
Service Info: OS: Unix  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 1.90 seconds  
  
kali@kali: ~  
$ sudo docker port juice-shop
```

- v. **Web Service Identification :** nmap -p 80,443,3000 --script=http-title,http-server-header 192.168.56.103



Findings

The enumeration surfaces the server's open ports, active services, version data, and web-facing interfaces. The web service matters most here since it is the direct entry point into the Juice Shop application.

Security Analysis

- The services uncovered hand an attacker a head start on reconnaissance - versions can be cross-referenced against known CVEs and advisories.
- The exposed web service itself warrants deeper testing for issues like broken authentication, injection flaws, broken access control, cross-site scripting, and misconfiguration.
- Administrative services such as SSH widen the attack surface whenever they're exposed unnecessarily or left poorly configured.
- Any development or auxiliary services running on non-standard ports should be reviewed to confirm they're actually required.

Recommendations

- Close off ports and services that serve no purpose.
- Limit administrative access, such as SSH, to trusted networks only.
- Keep the OS and application dependencies patched.
- Serve the application over HTTPS with a secure TLS configuration.
- Strip unnecessary server and version details from HTTP response headers.
- Apply firewall rules and segment the network appropriately.
- Re-scan the server periodically to catch newly exposed services.

Conclusion

The Nmap enumeration gives a network-level snapshot of the Juice Shop server, mapping out its attack surface and setting the stage for the next phase of testing. The exposed web service deserves the closest attention since it is the direct gateway into the application, while administrative and support services that aren't required should be locked down.

5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Objective :

This engagement enumerates the gateway device at 192.168.56.1 using Nmap - identifying open ports and services, spotting management interfaces, fingerprinting the OS, running script-based checks, weighing the security implications, and recommending controls to shrink the attack surface.

1. Host Discovery : `sudo nmap -sn 192.168.56.1`

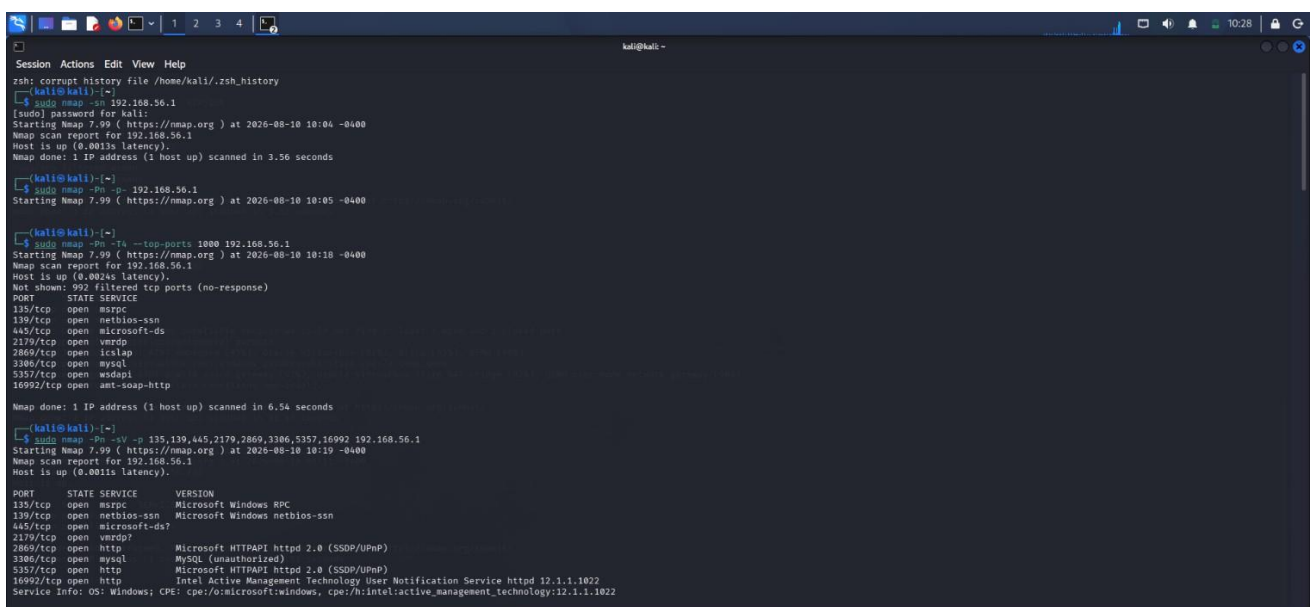
Purpose: Confirms whether the target host is up and reachable.

Result: The host at 192.168.56.1 responded as up.

2. TCP Port Scan : `sudo nmap -Pn -T4 --top-ports 1000 192.168.56.1`

Purpose: Sweeps the 1,000 most commonly used TCP ports to flag exposed services.

Result: Eight TCP ports came back open - 135, 139, 445, 2179, 2869, 3306, 5357, and 16992.



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ sudo nmap -sn 192.168.56.1
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:04 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0013s latency).
Nmap done: 1 IP address (1 host up) scanned in 3.56 seconds

kali@kali:~$ sudo nmap -Pn -p- 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:05 -0400

kali@kali:~$ sudo nmap -Pn -T4 --top-ports 1000 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:10 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0024s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
2179/tcp   open  vncop
2869/tcp   open  icslap
3306/tcp   open  mysql
5357/tcp   open  wsdapi
16992/tcp  open  ant-soap-http

Nmap done: 1 IP address (1 host up) scanned in 6.54 seconds

kali@kali:~$ sudo nmap -Pn -v -o 135,139,445,2179,2869,3306,5357,16992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0011s latency).

PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
2179/tcp    open  vncop?
2869/tcp    open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
3306/tcp    open  mysql        MySQL (unauthorized)
5357/tcp    open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
16992/tcp  open  http         Intel Active Management Technology User Notification Service httpd 12.1.1.1022
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022
```

3. Service and Version Detection : `sudo nmap -Pn -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1`

Purpose: Pulls service and version details for the ports found open.

4. UDP Port Scan : `sudo nmap -Pn -sU --top-ports 100 192.168.56.1`

Purpose: Flags commonly exposed UDP services such as NTP, NetBIOS, IKE, and UPnP.

Result: UDP port 1900 (UPnP) came back open, with a handful of others reported open/filtered.

5. Operating System Detection : `sudo nmap -Pn -O 192.168.56.1`

Purpose: Attempts to fingerprint the OS and device type.

Result: OS fingerprinting was inconclusive - Nmap pointed to a possible gateway/bridge/virtualized device without confirming an exact match.

```

Session Actions Edit View Help
1337/tcp open wdsapi
18992/tcp open ms-ssnp-http

Nmap done: 1 IP address (1 host up) scanned in 8.54 seconds

---(Nmap Scan)---
# sudo nmap -sC -sV -o - 135,139,443,2179,2869,3386,3357,16992 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:19 -0400
Nmap scan report for 192.168.56.1
Host is up (0.001s latency).

PORT      STATE SERVICE      VERSION
135/tcp    open  wsdapi       Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
443/tcp    open  https        Microsoft HTTPAPI httpd 2.0 (SSRP/PHP)
2179/tcp   open  vncserver?   Microsoft HTTPAPI httpd 2.0 (SSRP/PHP)
2869/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSRP/PHP)
3357/tcp   open  mysql        MySQL (unauthorized)
3386/tcp   open  http         Microsoft HTTPAPI httpd 2.0 (SSRP/PHP)
18992/tcp   open  http         Intel Active Management Technology Httpd 12.1.1.1602
Service Info: OS: Windows; CPE: cpe:/o:microsoft/windows, cpe:/h:intel/active_management_technology/12.1.1.1602

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 23.58 seconds

---(Nmap Scan)---
# sudo nmap -sC -sV -o - --top-ports 100 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.071s latency).
Not shown: 92 filtered out ports (port-unreach)

PORT      STATE SERVICE
123/udp    open/filtered ntp
137/udp    open/filtered netbios-ns
138/udp    open/filtered netbios-dgm
140/udp    open/filtered lispdm
1588/udp    open         upnp
4788/udp    open/filtered nat-t-like
1337/udp    open/filtered saratool

Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds

---(Nmap Scan)---
# sudo nmap -sC -sV -o - 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2020-08-18 18:21 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0001s latency).
Not shown: 992 filtered tcp ports (no-response)

PORT      STATE SERVICE

```

6. Default NSE Script Enumeration : `sudo nmap -Pn -sC -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1`

Purpose: Combines Nmap's default scripts with service detection to pull extra detail on the exposed services.

Result: Returned SMB security settings, HTTP details, and Intel AMT information.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 3.58 seconds  
kali@kali:~$ sudo nmap -Pn -O 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:21 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00095s latency).  
Not shown: 992 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
2179/tcp  open  vmrpd  
2869/tcp  open  icslap  
3306/tcp  open  mysql  
5357/tcp  open  wsddapi  
16992/tcp open  amt-soap-http  
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T B0d210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 10.70 seconds  
kali@kali:~$ sudo nmap -Pn -sC -sV -p 135,139,445,2179,2869,3306,5357,16992 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:22 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0036s latency).  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
2179/tcp  open  vmrpd?  
2869/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
3306/tcp  open  mysql        MySQL (unauthorized)  
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)  
|_ http-server-header: Microsoft-HTTPAPI/2.0  
|_ http-title: Service Unavailable  
16992/tcp open  http         Intel Active Management Technology User Notification Service httpd 12.1.1.1022  
|_ http-server-header: Intel(R) Active Management Technology 12.1.1.1022  
|_ http-title: IntelSreg; Active Management Technology  
|_ Requested resource was /login.htm  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows, cpe:/h:intel:active_management_technology:12.1.1.1022
```

7. Management Port Check : `sudo nmap -Pn -sV -p 22,23,80,443,161,162 192.168.56.1`

Purpose: Checks the usual management services - SSH, Telnet, HTTP, HTTPS, and SNMP.

Result: Every management port checked came back closed.

8. SMB Security Enumeration : `sudo nmap -Pn -p 139,445 --script smb-protocols,smb2-security-mode,smb2-time 192.168.56.1`

Purpose: Confirms SMB protocol versions and reviews SMB security configuration.

Result: SMB 3.1.1 was in use with message signing enabled and enforced.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 8.98 seconds  
kali@kali:~$ sudo nmap -Pn -p 2869,5357 --script http-title,http-headers,http-methods 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:26 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00092s latency).  
PORT      STATE SERVICE  
2869/tcp  open  icslap  
5357/tcp  open  wsddapi  
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds  
kali@kali:~$ sudo nmap -Pn -sU -p 1900 --script upnp-info 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:26 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.81s latency).  
PORT      STATE SERVICE  
1900/udp  open  upnp  
|_ upnp-info:  
|   uuid:8651b45e-e605-4600-a665-cb16ea8d3512  
|   location: http://192.168.56.1:2869/upnp/udhisapi.dll?content=uuid:8651b45e-e605-4600-a665-cb16ea8d3512  
|   server: Microsoft-Windows/10.0 UPnP/1.0 UPnP-Device-Host/1.0  
|   webserver: Microsoft-Windows/10.0 UPnP/1.0 UPnP-Device-Host/1.0 Microsoft-HTTPAPI/2.0  
|   deviceType: urn:schemas-upnp-org:device:MediaRenderer:1  
|   friendlyName: Swetha  
|   modelName: Windows Digital Media Renderer  
|   modelDescription: Digital Media Renderer  
|   manufacturer: Microsoft Corporation  
|   services:  
|   |_ urn:schemas-upnp-org:service:RenderingControl:1  
|   |_ urn:schemas-upnp-org:service:AVTransport:1  
|   |_ urn:schemas-upnp-org:service:ConnectionManager:1  
Nmap done: 1 IP address (1 host up) scanned in 3.84 seconds  
kali@kali:~$ sudo nmap -Pn -p 16992 --script http-title,http-headers,http-methods 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:27 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.0013s latency).  
PORT      STATE SERVICE  
16992/tcp open  amt-soap-http  
Nmap done: 1 IP address (1 host up) scanned in 0.62 seconds
```

```
Session Actions Edit View Help
| smb2-security-mode:
| 3.1.1:
| _ Message signing enabled and required
| smb2-time:
| _ date: 2026-08-10T14:24:14
| _ start_date: N/A
| _ clock-skew: 1m86s
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.94 seconds

(kali@kali)-[~]
└─$ sudo nmap -Pn -v -o 22,23,80,443,161,162 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:24 -0400
Nmap scan report for 192.168.56.1
Host is up (2.8s latency).

PORT      STATE SERVICE VERSION
22/tcp    closed ssh
23/tcp    closed telnet
80/tcp    closed http
161/tcp   closed snmp
162/tcp   closed snmptrap
443/tcp   closed https

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.95 seconds

(kali@kali)-[~]
└─$ sudo nmap -Pn -v -o 139,445 --script smb-protocols,smb2-security-mode,smb2-time 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 10:24 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0059s latency).

PORT      STATE SERVICE
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds

Host script results:
| smb2-security-mode:
| 3.1.1:
| _ Message signing enabled and required
| smb2-time:
| _ date: 2026-08-10T14:26:07
| _ start_date: N/A
| smb-protocols:
| _ dialects:
| _ 2.0.2
| _ 2.1
```

Service Risk Analysis

- **135 - MSRPC:** supports Windows inter-process communication;
- **139 - NetBIOS:** enables network-share discovery, which can hand an attacker useful reconnaissance data.
- **445 - SMB:** the file-sharing service; if exposed, it can open the door to unauthorized access.
- **2179 - VMRDP:** not fully fingerprinted - worth investigating and disabling if unused.
- **2869/5357 - HTTPAPI/UPnP:** may leak system details if left open.
- **3306 - MySQL:** a database service that should be reachable only by authorized hosts.

Security Implications

These exposed services collectively broaden the gateway's attack surface. Anyone with network access could enumerate the services, attempt authentication attacks, or try to exploit a vulnerable one - SMB, MySQL, UPnP, and Intel AMT are the ones that need the most attention.

Recommended Security Controls

- Disable any service or port that isn't required.
- Lock SMB/NetBIOS down to trusted systems only.
- Restrict MySQL access to authorized hosts.
- Turn off UPnP unless it's genuinely needed.

Conclusion

The assessment turned up several exposed TCP and UDP services on 192.168.56.1. SSH, Telnet, HTTP, HTTPS, and SNMP were all closed, but SMB, MySQL, UPnP, and Intel AMT remained reachable.

Restricting the unnecessary services and layering in proper firewalling, authentication, patching, and network segmentation will bring down the gateway's overall risk.

Code of Conduct:

I J Sravani (192372141) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: J Sravani

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	